ML LAB 5

BL.SC.U4AIE24027

N Tarun

Code:

```
import sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import pandas as pd
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score,
davies_bouldin_score
import matplotlib.pyplot as plt

#1
path = r"c:\Users\ADMIN\Downloads\Blog_TFIDF_Vectors.xlsx"
df = pd.read_excel(path)

X_single = df.iloc[:, [0]]
y = df.iloc[:, 200]

X_train, X_test, y_train, y_test = train_test_split(
    X_single, y, test_size=0.2, random_state=42
)

reg = LinearRegression().fit(X_train, y_train)

y_train_pred = reg.predict(X_train)

print("Predicted values on training data:")
print(y_train_pred)

#2
X_single = df.iloc[:, 0:50]
y = df.iloc[:, 200]

X_train, X_test, y_train, y_test = train_test_split(
    X_single, y, test_size=0.2, random_state=42
```

```
)

reg = LinearRegression().fit(X_train, y_train)

y_train_pred = reg.predict(X_train)

print("Predicted values on training data:")
print(y_train_pred)

#3
y_train_pred = reg.predict(X_train)
y_test_pred = reg.predict(X_test)

mse_train = mean_squared_error(y_train, y_train_pred)
rmse_train = np.sqrt(mse_train)
mape_train = np.mean(np.abs((y_train - y_train_pred) / y_train)) * 100
r2_train = r2_score(y_train, y_train_pred)

mse_test = mean_squared_error(y_test, y_test_pred)
rmse_test = np.sqrt(mse_test)
mape_test = np.mean(np.abs((y_test - y_test_pred) / y_test)) * 100
r2_test = r2_score(y_test, y_test_pred)

print("MSE:", mse_train)
print("RMSE:", rmse_train)
print("MAPE:", mape_train)
print("R2:", r2_train)

print("MSE:", mse_test)
print("RMSE:", rmse_test)
print("MAPE:", mape_test)
print("R2:", r2_test)

#4

X = df.drop(columns=['Author_Label'])

X_train, X_test = train_test_split(
    X, test_size=0.2, random_state=42
)

kmeans = KMeans(n_clusters=5, random_state=0, n_init="auto")
```

```python
kmeans.fit(X_train)

print("Cluster Labels:")
print(kmeans.labels_)

print("\nCluster Centers:")
print(kmeans.cluster_centers_)

#5
X = df.drop(columns=['Author_Label'])

X_train, X_test = train_test_split(
    X, test_size=0.2, random_state=42
)

kmeans = KMeans(n_clusters=2, random_state=42, n_init="auto")
kmeans.fit(X_train)

labels = kmeans.labels_

sil_score = silhouette_score(X_train, labels)
ch_score = calinski_harabasz_score(X_train, labels)
db_index = davies_bouldin_score(X_train, labels)

print("Silhouette Score:", sil_score)
print("Calinski-Harabasz Score:", ch_score)
print("Davies-Bouldin Index:", db_index)

#6
X = df.drop(columns=['Author_Label'])

k_values = range(2, 11)

sil_scores = []
ch_scores = []
db_scores = []

for k in k_values:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init="auto")
    kmeans.fit(X)
    labels = kmeans.labels_
```

```
    sil_scores.append(silhouette_score(X, labels))
    ch_scores.append(calinski_harabasz_score(X, labels))
    db_scores.append(davies_bouldin_score(X, labels))

plt.figure()
plt.plot(k_values, sil_scores, marker='o')
plt.xlabel("k")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Score vs k")
plt.show()

plt.figure()
plt.plot(k_values, ch_scores, marker='o')
plt.xlabel("k")
plt.ylabel("Calinski-Harabasz Score")
plt.title("CH Score vs k")
plt.show()

plt.figure()
plt.plot(k_values, db_scores, marker='o')
plt.xlabel("k")
plt.ylabel("Davies-Bouldin Index")
plt.title("DB Index vs k")
plt.show()

#7
X = df.drop(columns=['Author_Label'])

X_train, X_test = train_test_split(
    X, test_size=0.2, random_state=42
)

distortions = []

for k in range(2, 20):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init="auto")
    kmeans.fit(X_train)
    distortions.append(kmeans.inertia_)

plt.figure()
plt.plot(range(2, 20), distortions, marker='o')
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia")
```

```
plt.title("Elbow Plot")
plt.show()
```

Output:

Predicted values on training data: [2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 3.529019 2.90264506 4.22336993 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506 2.90264506] Predicted values on training data: [3. 2. 5. 4. 4.33333333 5. 2. 1. 4.33333333 4.33333333 4. 1. 4. 1. 4. 3. 2. 2.

1. 3.]

MSE: 0.13333333333333333 RMSE: 0.3651483716701107 MAPE: 3.555555555555666 R2: 0.9333333333333333 MSE: 3.8483872375661945 RMSE: 1.9617306740646623 MAPE: 92.9222776454399 R2: -0.9241936187830972 Cluster Labels: [3 2 3 2 4 2 2 0 4 4 0 1 0 1 3 4 2 2 1 3]

Cluster Centers: [[1.16071687e-01 1.73472348e-18 1.73472348e-18 0.00000000e+00 0.00000000e+00 6.78797614e-02 0.00000000e+00 3.46944695e-18 1.73472348e-18 0.00000000e+00 1.30993230e-01 1.73472348e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.55756954e-01 0.00000000e+00 1.96829718e-02 2.99387817e-02 1.73472348e-18 1.73472348e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.53647998e-02 0.00000000e+00 2.91479638e-02 1.56513493e-02 0.00000000e+00 0.00000000e+00 1.65006276e-02 2.33974528e-01 6.93889390e-18 7.61892668e-02 2.99387817e-02 1.15554312e-02 9.25442451e-03 3.71948780e-02 3.14716714e-02 0.00000000e+00 0.00000000e+00 3.30012552e-02 1.73472348e-18 0.00000000e+00 0.00000000e+00 8.67361738e-19 1.73472348e-18 3.46944695e-18 2.64466188e-02 1.21780234e-02 3.13026985e-02 3.46944695e-18 0.00000000e+00 1.61371760e-01 1.73472348e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00 3.11010333e-02 2.20932556e-02 0.00000000e+00 7.44515453e-02 0.00000000e+00 1.38777878e-17 6.95321505e-02 0.00000000e+00 1.86698999e-02 3.46944695e-18 2.25809198e-02 0.00000000e+00 1.73472348e-18 1.15886918e-02 1.73472348e-18 0.00000000e+00 3.46944695e-18 8.67361738e-19 3.55887374e-02 2.16024083e-01 3.46944695e-18 0.00000000e+00 1.72966086e-02 5.98775634e-02 0.00000000e+00 0.00000000e+00 1.19316459e-01 2.52595910e-02 1.15554312e-02 8.67361738e-19 8.06903927e-02 4.30493834e-02 4.00192153e-02 1.28388311e-02 8.77810429e-03

3.07295996e-02 3.46944695e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00
2.43649112e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 3.46944695e-18 3.46944695e-18 1.73472348e-18
0.00000000e+00 1.65006276e-02 1.28740223e-02 0.00000000e+00 5.09098210e-02
1.73472348e-18 1.38777878e-17 0.00000000e+00 3.46944695e-18 8.27198749e-02
2.64466188e-02 1.56513493e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 1.35725965e-02 1.73472348e-18 1.73472348e-18
4.33680869e-19 1.78907685e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.78907685e-02 1.36630808e-02 6.57672600e-02 0.00000000e+00 1.73472348e-18
1.65006276e-02 2.97697267e-01 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.17856034e-01 1.53647998e-02 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 1.44044711e-02 1.73472348e-18 0.00000000e+00
1.65006276e-02 1.73472348e-18 1.35725965e-02 1.65006276e-02 1.56513493e-02
0.00000000e+00 3.21519769e-02 5.09098210e-02 1.73472348e-18 4.17306326e-02
5.40975864e-02 3.65473669e-02 6.93889390e-18 1.36630808e-02 1.91343255e-02
1.36630808e-02 2.80675519e-02 1.73472348e-18 3.46944695e-18 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
4.45127636e-02 0.00000000e+00 1.73472348e-18 5.14960893e-02 2.00101573e-02
1.54474975e-01 0.00000000e+00 1.15886918e-02 2.73261615e-02 6.93889390e-18
1.73472348e-18 3.52933181e-02 0.00000000e+00 1.28740223e-02 3.46944695e-18
2.43560469e-02 8.67361738e-19 2.12809602e-01 7.75420602e-02 0.00000000e+00
1.73472348e-18] [0.00000000e+00 1.73472348e-18 1.73472348e-18 0.00000000e+00
0.00000000e+00 1.73472348e-18 0.00000000e+00 3.46944695e-18 1.73472348e-18
0.00000000e+00 2.79943949e-02 1.73472348e-18 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.73472348e-18 1.73472348e-18 8.20686623e-02 0.00000000e+00
1.09114806e-01 1.73472348e-18 0.00000000e+00 1.73472348e-18 0.00000000e+00
0.00000000e+00 0.00000000e+00 3.59787493e-02 0.00000000e+00 6.93889390e-18
3.46944695e-18 0.00000000e+00 1.03536394e-01 0.00000000e+00 0.00000000e+00
1.39105672e-01 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.73472348e-18
0.00000000e+00 0.00000000e+00 8.67361738e-19 1.73472348e-18 3.46944695e-18
1.73472348e-18 3.46944695e-18 8.67361738e-19 3.46944695e-18 0.00000000e+00
0.00000000e+00 1.73472348e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.73472348e-18 0.00000000e+00 0.00000000e+00 1.73472348e-18 6.51200199e-01
5.38159808e-01 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.22420885e-01
0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00 1.73472348e-18
0.00000000e+00 1.03536394e-01 8.67361738e-19 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 7.54417819e-02 1.73472348e-18 0.00000000e+00

0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 8.67361738e-19
8.79705753e-02 3.46944695e-18 0.00000000e+00 0.00000000e+00 1.19009180e-01
1.73472348e-18 3.46944695e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 3.46944695e-18 3.46944695e-18 1.73472348e-18
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.73472348e-18
1.73472348e-18 1.38777878e-17 0.00000000e+00 3.46944695e-18 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.52050269e-01 0.00000000e+00 0.00000000e+00 1.73472348e-18 1.73472348e-18
4.33680869e-19 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.73472348e-18
7.19574986e-02 6.93889390e-18 0.00000000e+00 0.00000000e+00 2.79943949e-02
0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00 0.00000000e+00
3.46944695e-18 0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00
0.00000000e+00 1.73472348e-18 1.15350939e-01 1.73472348e-18 8.67361738e-19
0.00000000e+00 0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00
7.86516095e-02 3.46944695e-18 6.93889390e-18 0.00000000e+00 0.00000000e+00
1.22420885e-01 3.46944695e-18 1.73472348e-18 3.46944695e-18 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.95943317e-02 0.00000000e+00
1.73472348e-18 0.00000000e+00 1.73472348e-18 0.00000000e+00 0.00000000e+00
6.93889390e-18 0.00000000e+00 0.00000000e+00 1.73472348e-18 6.93889390e-18
1.73472348e-18 6.93889390e-18 0.00000000e+00 0.00000000e+00 3.46944695e-18
1.38777878e-17 8.67361738e-19 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.73472348e-18] [0.00000000e+00 3.57636731e-02 1.99876000e-02 0.00000000e+00
8.90083578e-02 0.00000000e+00 2.16715105e-02 3.46944695e-18 0.00000000e+00
0.00000000e+00 7.77598972e-03 1.73472348e-18 0.00000000e+00 9.99380002e-03
4.77629404e-02 8.22040339e-03 2.49951683e-02 3.25929492e-02 0.00000000e+00
3.46944695e-18 0.00000000e+00 1.73472348e-18 4.69408308e-02 2.37554504e-02
7.77598972e-03 9.30587249e-03 3.47658819e-02 9.30587249e-03 2.99814000e-02
1.08357553e-02 4.45041789e-02 6.99566001e-02 9.30587249e-03 1.63945285e-01
3.46944695e-18 0.00000000e+00 6.93889390e-18 2.32835106e-01 7.77598972e-03
9.92305661e-02 8.72423805e-03 1.73472348e-18 9.99380002e-03 9.99380002e-03
0.00000000e+00 1.73472348e-18 8.67361738e-19 0.00000000e+00 0.00000000e+00
8.22040339e-03 6.06949397e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
5.23454283e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.08357553e-02
6.38850706e-03 3.66192555e-02 8.72423805e-03 1.73472348e-18 1.38777878e-17
0.00000000e+00 7.01882804e-03 0.00000000e+00 0.00000000e+00 0.00000000e+00
4.54350100e-02 4.49927696e-02 0.00000000e+00 5.24538381e-02 0.00000000e+00

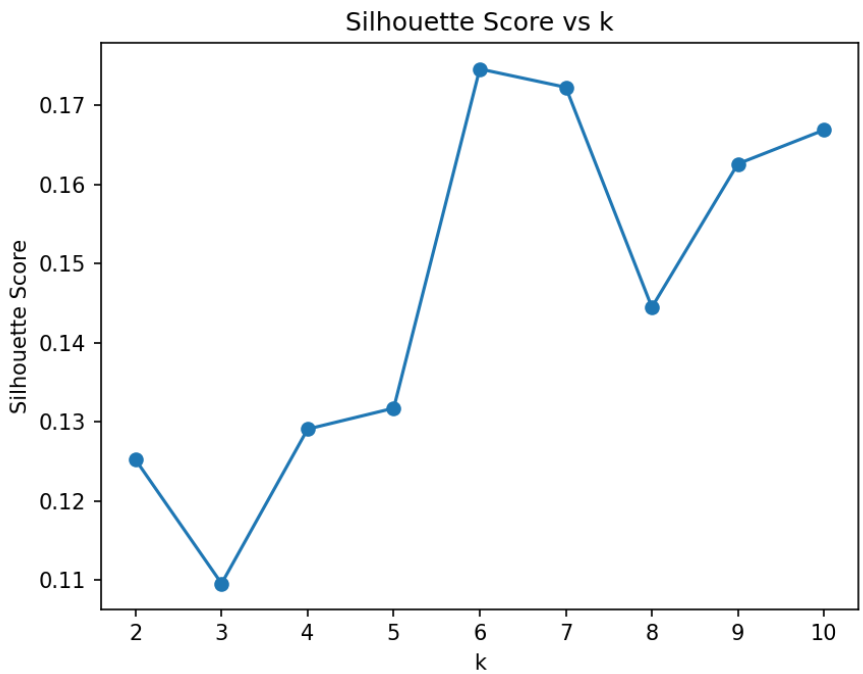
0.00000000e+00 3.46944695e-18 0.00000000e+00 2.22043303e-01 8.72423805e-03
7.37844814e-03 1.73472348e-18 3.82102619e-02 1.73472348e-18 0.00000000e+00
1.62671533e-01 6.49704050e-02 1.19917287e-01 3.03044659e-02 0.00000000e+00
6.93889390e-18 3.16856890e-02 0.00000000e+00 1.73472348e-18 0.00000000e+00
9.30587249e-03 1.99876000e-02 0.00000000e+00 6.73173158e-02 4.99690001e-02
4.77629404e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00 8.14854831e-02
1.05161729e-01 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
3.25929492e-02 0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00
0.00000000e+00 5.37913667e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 9.99380002e-03 1.49139953e-01 0.00000000e+00 1.73472348e-18
0.00000000e+00 0.00000000e+00 1.73472348e-18 1.73472348e-18 0.00000000e+00
1.19212244e-02 1.73472348e-18 0.00000000e+00 1.86117450e-02 1.38777878e-17
0.00000000e+00 8.72423805e-03 3.46944695e-18 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 1.08357553e-02 8.67361738e-19 7.27690111e-02
0.00000000e+00 0.00000000e+00 3.82207058e-02 5.75919569e-02 0.00000000e+00
0.00000000e+00 0.00000000e+00 1.73472348e-18 0.00000000e+00 1.08357553e-02
1.99876000e-02 0.00000000e+00 3.46944695e-18 9.99380002e-03 9.99380002e-03
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
3.92353606e-02 7.37844814e-03 1.55519794e-02 1.73472348e-18 1.22177724e-02
8.72423805e-03 0.00000000e+00 0.00000000e+00 0.00000000e+00 8.67361738e-19
8.67361738e-19 0.00000000e+00 3.89310728e-02 0.00000000e+00 9.51838494e-02
9.30587249e-03 7.37844814e-03 0.00000000e+00 1.73472348e-18 3.66438876e-02
2.98695582e-02 9.99380002e-03 1.73472348e-18 0.00000000e+00 6.52542534e-02
1.98500611e-02 1.47568963e-02 0.00000000e+00 0.00000000e+00 3.46944695e-18
0.00000000e+00 9.30587249e-03 4.99690001e-02 3.84161820e-02 4.10461341e-02
0.00000000e+00] [0.00000000e+00 0.00000000e+00 2.46550883e-02 1.95621116e-01
0.00000000e+00 1.31321441e-02 2.02538067e-02 7.87928649e-02 3.35437176e-02
5.81826333e-02 1.62377250e-02 3.99683534e-02 3.44887071e-02 6.77634601e-02
1.82029148e-02 7.71642901e-02 2.13290682e-02 6.28943450e-03 1.15328549e-01
9.75293143e-02 0.00000000e+00 6.87535203e-02 2.36435378e-02 0.00000000e+00
4.05235156e-02 3.97671740e-03 1.79867815e-02 3.97671740e-03 8.54138468e-03
1.31321441e-02 0.00000000e+00 0.00000000e+00 3.97671740e-03 0.00000000e+00
1.29317194e-02 3.35437176e-02 2.85699553e-02 1.40996289e-02 1.45346022e-02
6.83678456e-02 3.72816512e-03 1.34174870e-02 0.00000000e+00 4.99116042e-02
0.00000000e+00 2.25964686e-02 2.03773866e-02 0.00000000e+00 3.54475835e-02
1.34753880e-02 9.42393173e-03 0.00000000e+00 0.00000000e+00 0.00000000e+00
2.37633644e-02 4.27069234e-03 0.00000000e+00 0.00000000e+00 3.70439088e-02
7.88034608e-03 9.49694526e-02 3.83856042e-02 1.50045099e-02 0.00000000e+00

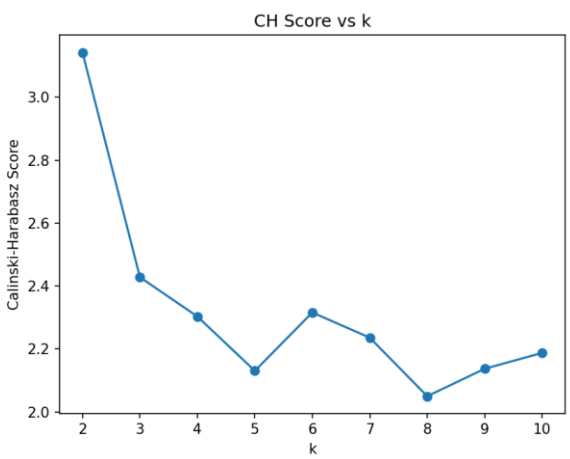
0.00000000e+00 2.66541699e-02 2.96708943e-02 0.00000000e+00 1.70509496e-02
5.05908992e-02 6.67744565e-02 4.59849428e-02 1.10587490e-02 4.27069234e-03
1.07615151e-02 3.15306309e-03 0.00000000e+00 5.94429578e-02 1.00175996e-02
6.99445503e-02 4.27418437e-02 2.85924100e-02 1.14789723e-02 1.33227845e-02
0.00000000e+00 1.12676286e-02 3.04658483e-02 5.94839432e-03 3.12466361e-02
5.35320121e-02 1.02170944e-02 1.67671914e-02 8.92877420e-03 7.89877762e-02
6.70874352e-03 7.33835121e-02 0.00000000e+00 8.05685181e-03 2.03843960e-02
1.06384686e-02 3.12196946e-02 0.00000000e+00 2.16140425e-02 2.19924504e-02
1.02754617e-01 3.97671740e-03 9.19698856e-02 1.37954828e-01 0.00000000e+00
6.28943450e-03 1.44093616e-02 6.62716602e-03 1.95790972e-02 2.18677677e-02
0.00000000e+00 1.24582165e-01 3.44887071e-02 9.19698856e-02 1.07615151e-02
2.75268254e-02 2.29507463e-02 4.63048860e-03 1.37457647e-02 1.73942044e-02
0.00000000e+00 1.00175996e-02 9.43907185e-03 4.62772946e-02 6.89774142e-02
0.00000000e+00 3.99822596e-02 5.73488260e-02 1.35040589e-01 1.42109985e-02
1.13563106e-01 1.07615151e-02 6.62716602e-03 2.55053097e-02 0.00000000e+00
8.54138468e-03 0.00000000e+00 3.70439088e-02 2.06531405e-02 4.40641119e-02
0.00000000e+00 1.21117559e-02 7.50225494e-03 1.74144378e-02 7.22382672e-02
0.00000000e+00 3.30894156e-02 6.28943450e-03 0.00000000e+00 1.38914658e-02
1.28120770e-02 0.00000000e+00 0.00000000e+00 2.51396212e-02 7.20468082e-03
0.00000000e+00 3.68755751e-02 8.73562351e-03 2.34379166e-02 4.26316679e-02
9.18300677e-02 2.22638291e-02 1.78575484e-02 1.25788690e-02 2.90608209e-02
0.00000000e+00 1.05731395e-02 0.00000000e+00 0.00000000e+00 2.87405892e-02
2.96298186e-02 2.64273192e-02 9.43907185e-03 4.28853946e-02 3.28334293e-02
1.92314727e-02 1.75737548e-02 4.92726935e-02 2.37929101e-02 4.63235026e-02
3.45754673e-02 2.41705554e-02 3.21610918e-02 3.72816512e-03 7.34530526e-02
1.60698241e-02 4.78731087e-02 4.63048860e-03 1.25533785e-02 2.66455690e-02
2.93552568e-02 1.19301522e-02 0.00000000e+00 1.98232550e-02 0.00000000e+00
0.00000000e+00] [0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.77895341e-02 2.89320088e-02
0.00000000e+00 1.99424153e-02 2.71237042e-02 0.00000000e+00 0.00000000e+00
4.18625104e-02 0.00000000e+00 8.57930280e-03 2.71237042e-02 0.00000000e+00
0.00000000e+00 6.73768456e-02 0.00000000e+00 1.09099332e-01 2.89320088e-02
0.00000000e+00 2.38659747e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
5.01810337e-02 0.00000000e+00 4.58792718e-02 2.46135340e-02 9.97120765e-03
6.16408074e-02 2.71237042e-02 2.89320088e-02 1.28151218e-02 0.00000000e+00
2.69507382e-01 2.71237042e-02 1.52866719e-02 6.73768456e-02 9.25582658e-02
0.00000000e+00 9.97120765e-03 0.00000000e+00 1.01065268e-01 5.78640176e-02

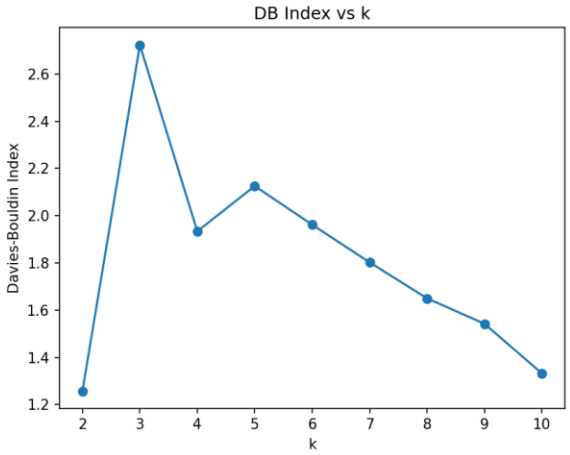
0.00000000e+00 6.21415586e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.98619036e-02 5.51180766e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 5.26434439e-02 1.28151218e-02 0.00000000e+00 0.00000000e+00
0.00000000e+00 2.55572794e-02 0.00000000e+00 2.40037325e-01 6.21415586e-02
1.11871533e-02 5.53407090e-02 3.05733438e-02 7.18739795e-03 0.00000000e+00
5.53407090e-02 2.89320088e-02 2.11440895e-01 0.00000000e+00 7.66217150e-02
0.00000000e+00 0.00000000e+00 5.41121517e-02 2.29396359e-02 0.00000000e+00
1.89925608e-02 2.08008654e-02 0.00000000e+00 2.41755949e-02 3.18009319e-02
1.19329874e-02 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
2.29396359e-02 2.71237042e-02 0.00000000e+00 3.10707793e-02 1.05410825e-02
4.16017309e-02 2.89320088e-02 0.00000000e+00 0.00000000e+00 6.73768456e-02
5.42474083e-02 0.00000000e+00 2.55572794e-02 2.55572794e-02 0.00000000e+00
6.73768456e-02 2.34783019e-01 0.00000000e+00 0.00000000e+00 0.00000000e+00
1.05410825e-02 0.00000000e+00 0.00000000e+00 2.71237042e-02 2.89320088e-02
4.16843012e-02 2.71237042e-02 2.55572794e-02 1.28151218e-02 0.00000000e+00
0.00000000e+00 0.00000000e+00 5.42474083e-02 2.89320088e-02 3.18252097e-01
0.00000000e+00 0.00000000e+00 2.55572794e-02 7.21967238e-02 6.73768456e-02
0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00 0.00000000e+00
0.00000000e+00 0.00000000e+00 0.00000000e+00 2.29396359e-02 0.00000000e+00
1.01065268e-01 0.00000000e+00 2.71237042e-02 6.73768456e-02 0.00000000e+00
0.00000000e+00 7.49566804e-02 2.55572794e-02 0.00000000e+00 0.00000000e+00
1.68442114e-01 0.00000000e+00 0.00000000e+00 5.11145588e-02 2.23743067e-02
9.43180780e-02 7.42635835e-02 1.94909607e-01 2.71237042e-02 1.89925608e-02
0.00000000e+00 5.42474083e-02 6.73768456e-02 1.01065268e-01 0.00000000e+00
0.00000000e+00 5.11145588e-02 2.10821650e-02 7.66718382e-02 0.00000000e+00
0.00000000e+00 4.18625104e-02 0.00000000e+00 0.00000000e+00 1.63840573e-02
3.63664439e-02 0.00000000e+00 3.08218688e-02 3.83108575e-02 5.61078645e-02
2.41755949e-02 6.07853848e-02 0.00000000e+00 2.54279387e-01 9.89960216e-02
2.74679269e-01 0.00000000e+00 0.00000000e+00 2.18215750e-02 9.32123379e-02
6.73768456e-02]] Silhouette Score: 0.05483848435803092 Calinski-Harabasz Score:
1.5387889610474108 Davies-Bouldin Index: 2.6395167372243367
C:\Users\ADMIN\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5
n2kfra8p0\LocalCache\local-packages\Python311\site-packages\sklearn\base.py:1365:
ConvergenceWarning: Number of distinct clusters (17) found smaller than n_clusters (18).
Possibly due to duplicate points in X. return fit_method(estimator, *args, **kwargs)
C:\Users\ADMIN\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5
n2kfra8p0\LocalCache\local-packages\Python311\site-packages\sklearn\base.py:1365:
ConvergenceWarning: Number of distinct clusters (17) found smaller than n_clusters (19).

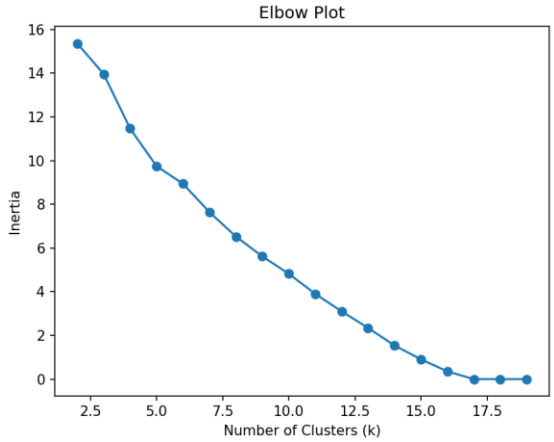
Possibly due to duplicate points in X. return fit_method(estimator, *args, **kwargs) PS C:\Users\ADMIN\AppData\Local\Programs\Microsoft VS Code>

Graphs:






DB Index vs k
Davies-Bouldin Index
k
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
2
3
4
5
6
7
8
9
10


Elbow Plot
Inertia
Number of Clusters (k)
16
14
12
10
8
6
4
2
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5